

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTMPE405B - SHEET METAL ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 Which of the following materials is commonly used in sheet metal fabrication?
a) Zinc b) Aluminium & Steel c) Lead d) None of the options
- Galvanized steel is primarily coated with which material to prevent corrosion.
a) Zinc b) Silicon c) Aluminium d) None of the options
- What type of process would you use to produce small holes in a sheet metal part?
a) Blanking b) Piercing c) Drawing d) None of the options
- The sheet metal operation performed to make the cutout at edge of the sheet is known as
a) Perforating b) Notching c) Shaving d) Piercing
- 5 In deep drawing, the common defect caused by insufficient blank holding force is
a) Tearing b) Wrinkling c) Surface Scratches d) None of the options
- 6 Formability of the material is determined by the following methods
a) Cupping Test b) Formability Diagram c) Both options a & b d) None of the options
- Which of the following dies performs two or more operations in a single stroke of the press?
a) Simple die b) Progressive Die c) Compound Die d) None of the options
- Which type of press is powered by a motor and uses a flywheel to store energy?
a) Hydraulic Press b) Pneumatic Press c) Mechanical Press d) None of the options
- 9 In aerospace applications, lightweight sheet metal components having high strength are typically made from:
a) Aluminium b) Stainless steel c) Titanium d) None of the options
- 10 Following type of Stainless Steel is preferred for sheet metal work for good corrosion resistance.
a) Ferritic Stainless Steel b) Martensitic Stainless Steel c) Austenetic Stainless Steel d) None of the options

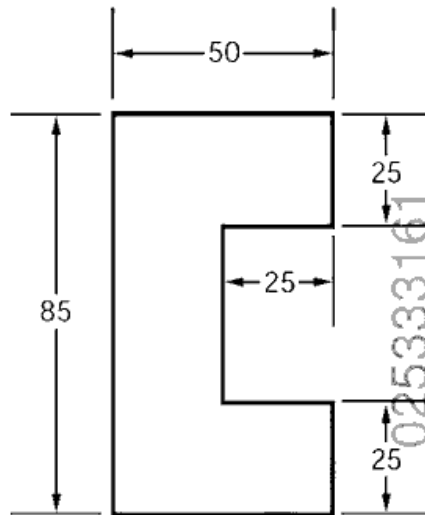
- 11 The sheet metal operation used to make pattern of holes in the sheet is known as
a) Shaving b) Trimming c) Perforating d) Notching
- 12 The shear is provided on the punch/ die for
a) Better strength of punch or die b) Reduction in shear force c) Increase in the shear force d) None of the options

Q2. Solve the following.

- A) Enlist any Five material used for sheet metal work and compare any two materials in terms of their material properties.
- B) Enlist any Five punching operations and explain any two of them with neat sketches.

Q3. Solve the following.

Determine the blanking force required to blank the following component through a sheet having a thickness of 1.25mm. The maximum shear strength of the material is 400 N/mm². All dimensions are in mm.



- B) A blanking operation is to be performed on 2.0 mm thick cold rolled steel. The shear strength of the material is 300 MPa. The part is circular with diameter = 75.0 mm. Determine the clearance and appropriate punch & die sizes for this operation.

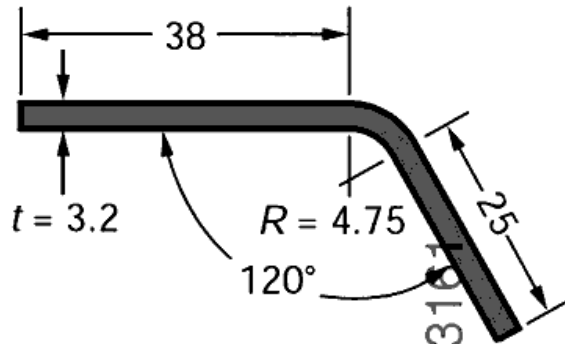
Q4. Solve Any Two of the following.

- A) A hole with 100mm diameter is to be punched in a steel plate of 6mm thickness. The material is cold rolled C40 steel for which the maximum shear strength can be taken as 550 Mpa. With normal clearance on the tool, cutting is complete at 40% penetration of the punch. Decide the shear and shear angle on the punch in order to bring the work within the capacity of a 200 KN press available in the shop.
- B) What is the importance of Stock Strip Layout? Determine the % utilization of the material for manufacturing a rectangular blank of size 60 mm x 20 mm from the long steel strip having the width of 70 mm. The advance distance is 25 mm.
- C) Explain with neat sketches the following.
a) V-Bending b) Bottoming c) Wipe Bending

Q5. Solve Any Two of the following.

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- A) Determine the bedding allowance & starting blank size for producing the following part. All dimensions are in mm.



- B) What is Springback? Why it Occurs? How the Springback is compensated during the bending operation?
- C) Differentiate between Progressive Die, Compound Die & Combination Die. Explain any one of it with neat sketch.

Q6. Solve Any Two of the following.

- A) A cup of 12 mm height and 6 mm diameter is to be made from a sheet metal of 2.6 mm thickness. Determine the diameter of cup after third draw reduction. Assume successive possible reductions of 40%, 30% and 20% in 1st, 2nd & 3rd draw.
- B) Compare Mechanical press with Hydraulic press stating their advantages, disadvantages & limitations.
- C) Describe in brief any one case study of sheet metal manufacturing of any product highlighting the following.
- a) Generic name of the product b) Material used c) Type of sheet metal operation involved d) Sequence of operations

*** End ***